

Diskrete Mathematik – Woche 4

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<https://github.com/tschai-yim/eth-resources>

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Predicate Logic

Problem 1 ★

Let $U = \mathbb{N}$, with the usual meaning of $+$, $\cdot$, $<$, $\leq$, $=$.

1. Determine the truth value of each of the following formulas:

$$F_1 = \forall x((x \geq 5) \rightarrow (x^2 \geq 25)), \quad F_2 = \forall x((x \geq 5) \wedge (x^2 \geq 25)),$$

$$F_3 = \exists x((x \geq 5) \wedge (x^2 \geq 25)), \quad F_4 = \exists x((x \geq 5) \rightarrow (x^2 \geq 25)).$$

2. Which of them formalize “every natural number ≥ 5 has a square ≥ 25 ”, and which formalize “some natural number ≥ 5 has a square ≥ 25 ”?
3. Determine the truth value of F_4 when $x^2 \geq 25$ is replaced by $x^2 < 0$. What does this say about using $\rightarrow$ under $\exists$?

Problem 2 ★ ★

Decide for each of the following formulas for which of the universes $U = \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ it is true ($+$, $\cdot$, $<$, $\leq$, $=$ with their usual meaning).

1. $G_1 = \forall x \exists y (y + y = x)$
2. $G_2 = \forall x ((x > 0) \rightarrow \exists y (y \cdot y = x))$
3. $G_3 = \forall x \exists y ((x < y) \wedge \forall z (\neg((x < z) \wedge (z < y))))$
4. $G_4 = \exists x \forall y (x \leq y)$

Problem 3 ★ ★

Let $U = \mathbb{N} \setminus \{0\}$. The only symbols you may use are the binary predicate $D(x, y) = 1 \Leftrightarrow$ “ x divides y ”, equality $=$, and the logical operators and quantifiers. Express the following statements as formulas (with free variables x, y, z as indicated):

1. “ x is prime”
2. “ x and y are coprime”, i.e. their only common divisor is 1
3. “ z is the greatest common divisor of x and y ” (★ ★ ★)

Problem 4 ★ ★ ★

As in the script (Sections 2.4.6–2.4.7), an informal justification suffices here; interpretations are only defined rigorously in Chapter 6.

1. Prove $\exists y \forall x (P(x, y)) \models \forall x \exists y (P(x, y))$, and disprove $\forall x \exists y (P(x, y)) \models \exists y \forall x (P(x, y))$ by giving an interpretation.
2. Is $\exists x (P(x) \rightarrow \forall y (P(y)))$ valid, i.e. true under every interpretation? Prove it or give an interpretation in which it is false.
3. Is $\exists y \forall x (Q(x, y) \leftrightarrow \neg Q(x, x))$ satisfiable?

Proof Techniques

Problem 5 ★ ★

In the script every proof pattern is justified by a lemma of propositional logic (Lemmas 2.5–2.9), where A stands for “ S is true” and B for “ T is true”. For each of the following patterns, write down the corresponding statement about formulas and then **prove** or **disprove** it.

1. To prove $S \Rightarrow T$: assume S and that T is false, and derive a contradiction.
2. To prove $S \Rightarrow T$: prove $\neg S \Rightarrow \neg T$.
3. To prove S : find a statement T and prove both $T \Rightarrow S$ and $\neg T \Rightarrow S$.

Problem 6 ★ ★

Each of the following “proofs” is flawed. Identify the invalid or unjustified step, and state whether the claim itself is true or false.

1. **Claim:** For all $n \in \mathbb{N}$: if n^2 is even, then n is even.
“Proof”: Let n be even, i.e. $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$ is even. ■
2. **Claim:** For all $x \in \mathbb{R}$: $x^2 \geq x$.
“Proof”: Case $x \leq 0$: then $x^2 \geq 0 \geq x$. Case $x \geq 1$: then $x^2 = x \cdot x \geq 1 \cdot x = x$. Both cases hold. ■
3. **Claim:** There are infinitely many primes.
“Proof”: Assume $p_1, \dots, p_n$ are all primes. Set $P = p_1 \cdot \dots \cdot p_n + 1$. No p_i divides P , hence P is prime. But $P > p_i$ for all i , contradicting that $p_1, \dots, p_n$ are all primes. ■

Problem 7 ★ ★

1. Prove indirectly (Definition 2.14): for all $x, y \in \mathbb{R}$, if $x \cdot y$ is irrational, then x is irrational or y is irrational.
2. Decide whether the converse holds: if x is irrational or y is irrational, then $x \cdot y$ is irrational.

Problem 8 ★ ★

1. Show by contradiction (Definition 2.17) that there is no smallest positive rational number, i.e. that for $U = \mathbb{Q}$

$$\neg \exists q (q > 0 \wedge \forall r ((r > 0) \rightarrow (q \leq r))).$$

2. The analogous statement is false for $U = \mathbb{N}$. Which step of your proof breaks, and which property of $\mathbb{Q}$ does the proof really use?

Problem 9 ★ ★ ★

Show that there exist irrational numbers a, b such that a^b is rational. You may use that $\sqrt{2}$ is irrational (Example 2.28) and that $(x^y)^z = x^{y \cdot z}$ for $x > 0$. Afterwards decide whether your proof is constructive or non-constructive (Definition 2.18).